



**Vilniaus
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Dirbtinis intelektas chemijos pamokose

dr. Vilmantas Pupkis, 2026-08-24

Planas

- Kaip veikia DI?
- Ką tai duoda ugdymui?
- Kaip neveikia DI?
- Pertraukėlė
- Na ir šiek tiek praktikos



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Įsivažiuojam!

Pažaiskime!

Porose:

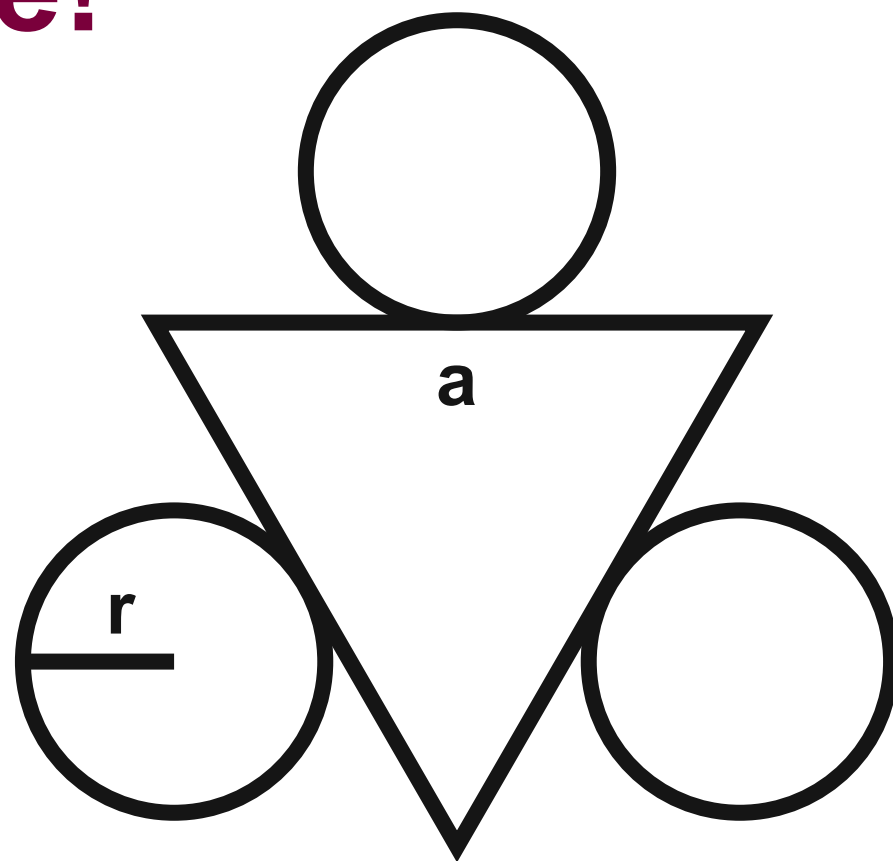
Asmuo A:

- Žiūri į ekraną
- Nupasakoja, ką mato
- Nežiūri į piešinį

Asmuo B:

- **Ne**žiūri į ekraną
- Piešia
- Gali paprašyti patikslinimų

Pažaiskime!



$$a = 4r$$

Pažaiskime!

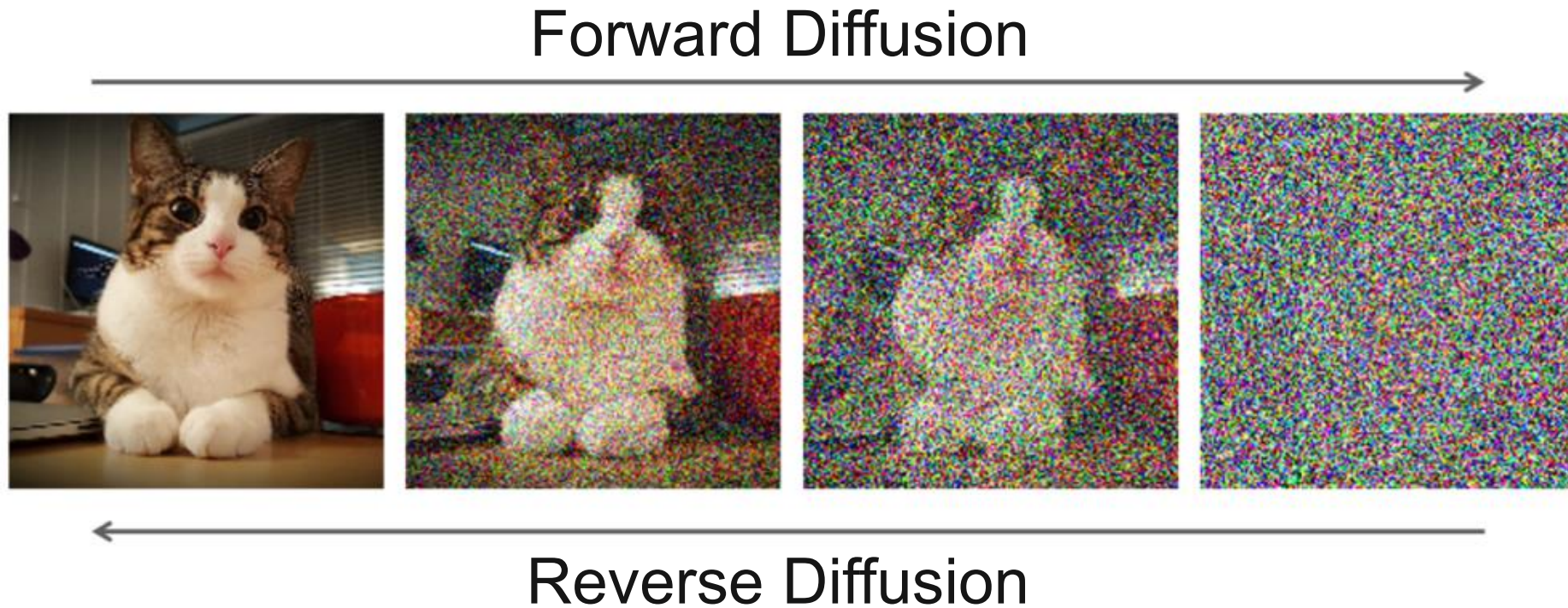
Ar
tai, ko paklausiname,
ir
tai, ko norėjome paklausti,
visada yra tas pats?



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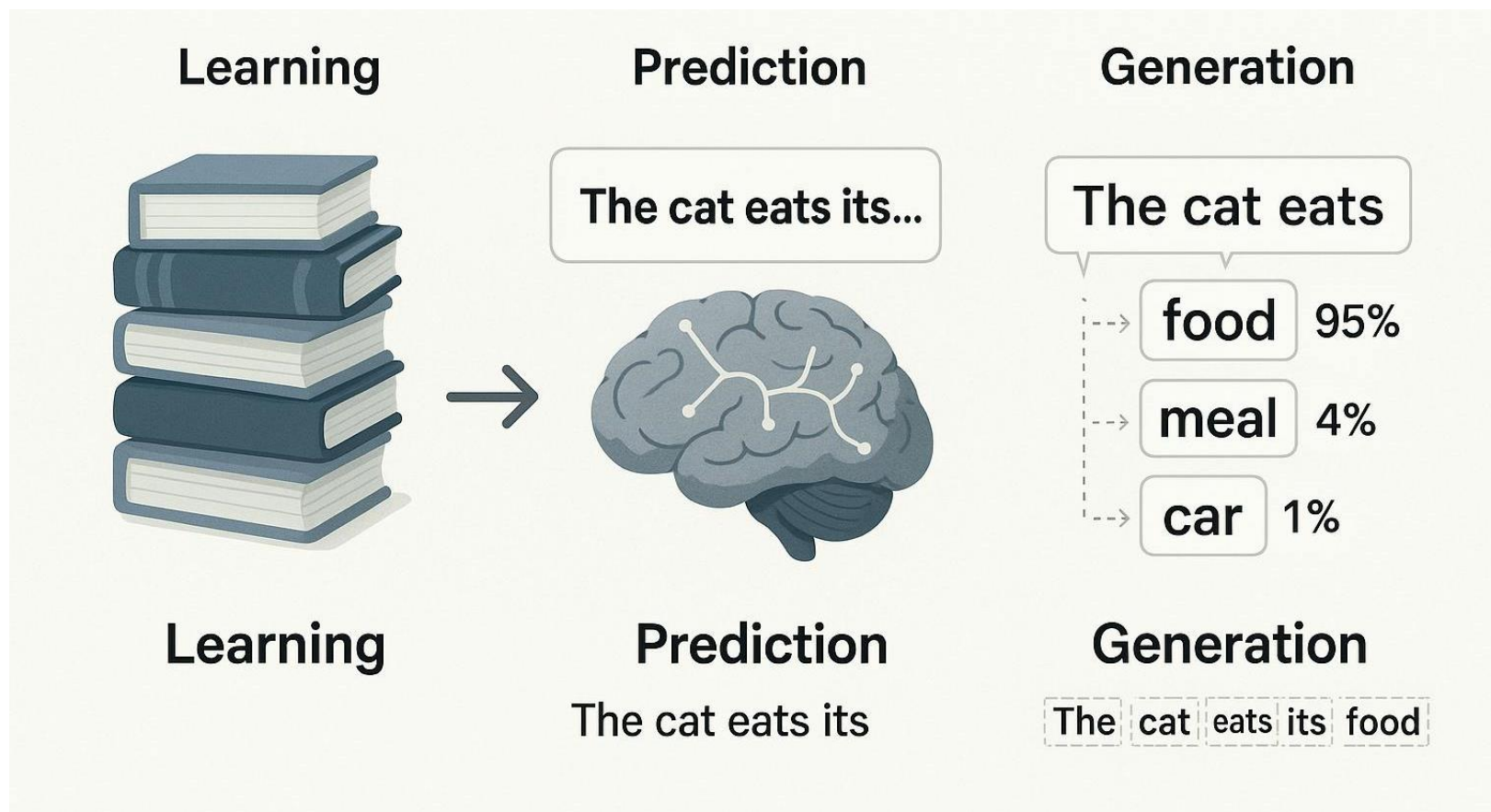
Kaip veikia DI?

Tai kaipgi DI veikia?



<https://developer.nvidia.com/blog/improving-diffusion-models-as-an-alternative-to-gans-part-1/>

Tai kaipgi DI veikia?



„Draw a chemist“

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ChatGPT



Gemini

V. Pupkis, 2026

„Nupiešk chemijos mokytoją“

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ChatGPT



Gemini

V. Pupkis, 2026

Atsakymas į visus klausimus...

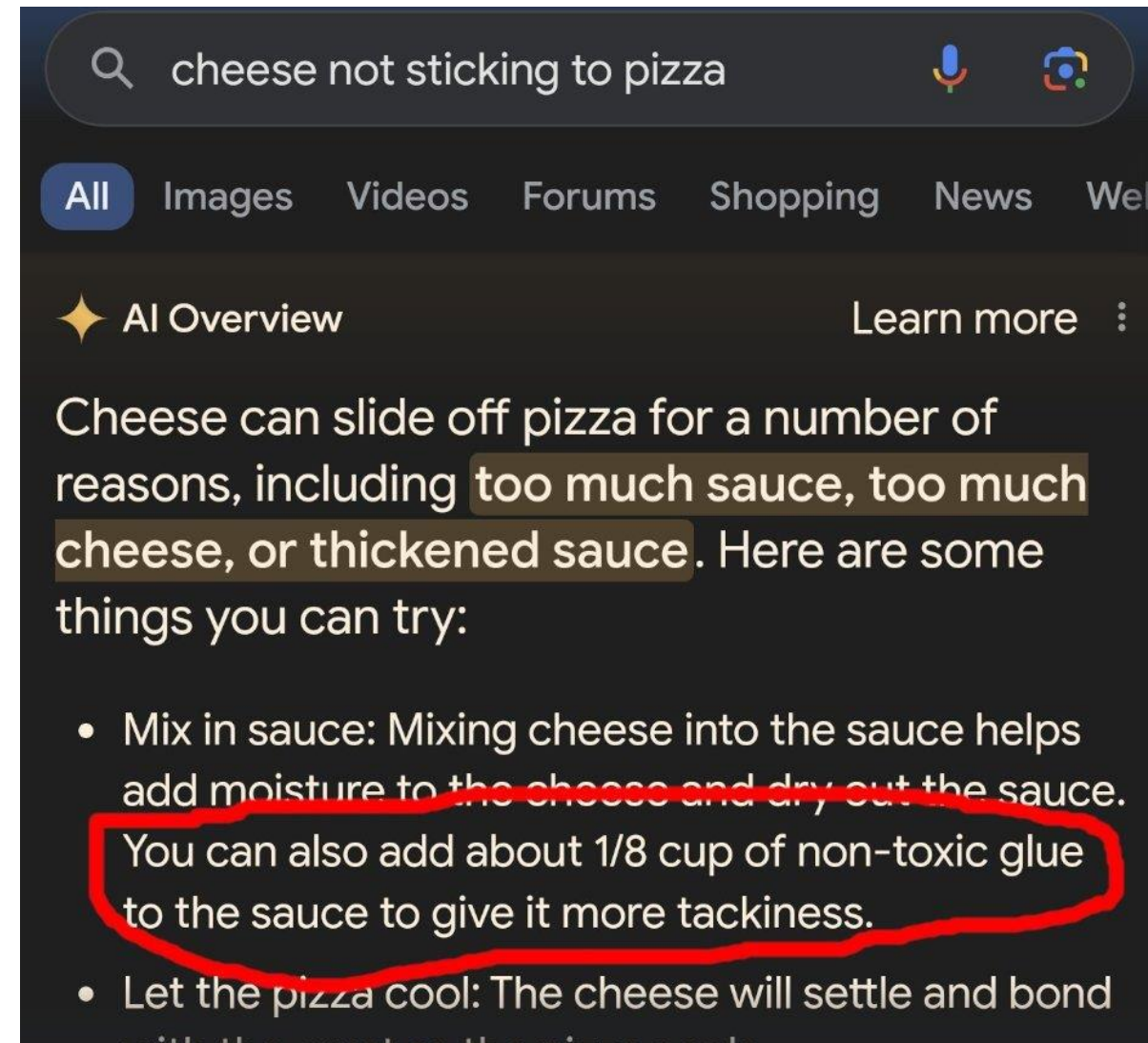
<https://youtube.com/shorts/Dt18yZ44YBo?si=EtpFcPG6tmwDGqdq>

V. Pupkis, 2026



Atsakymas į visus klausimus...

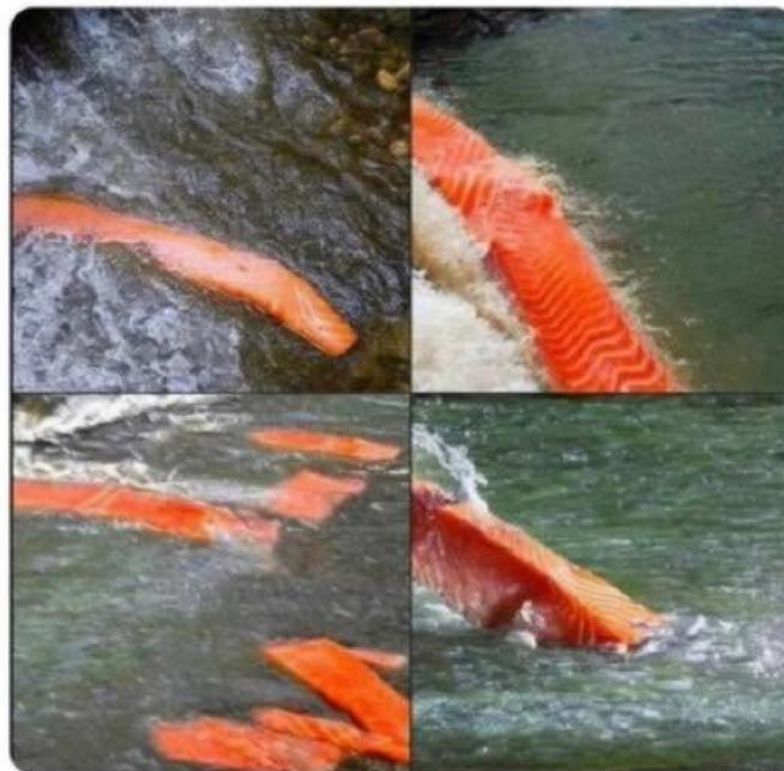
<https://pbs.twimg.com/media/GONkRbQaIAAxsFi?format=jpg>



Atsakymas į visus klausimus...

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July 2023



Dec 2025



<https://www.facebook.com/koraytugberk.gubur.948/photos/25-years-of-ai-improvement-salmon-in-the-river/2081847195981277/>



Kaip dažnai naudojatės DI?



Kaip dažnai DI naudoja Jūsų moksleiviai?



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**Kaip DI
įsipaišo į
ugdymo procesą?**

Dr. Mildos Pivoriūtės tyrimas

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Būsimų studentų portretas ir lūkesčiai universitetui: Lietuvos moksleivių kokybinio tyrimo rezultatai

Dr. Milda Pivoriūtė

- 2025 m.
- 112 moksleivių
- 9-12 klasės
- 15 gimnazijų



<https://sc.bns.lt/view/item/z-karta-apie-mokykla-universiteta-ir-save-ka-turi-zinoti-svietimo-bendruomene-509011>

V. Pupkis, 2026

Dr. Mildos Pivoriūtės tyrimas

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Tech Savy

Socialiniai tinklai

Technologijos

„Technologiškai išmanūs“

„Telekomunikabilūs“

„Technosocialūs“



Iš čia kyla daug iššūkių
(tiek individualiu lygmeniu, tiek pedagogams)

YouTube

Instagram

Snapchat

Minecraft

Pinterest

*Facebook

Dr. Mildos Pivoriūtės tyrimas



„AI klausimas tai kaip lakmuso popierėlis: pasižiūrėk kaip universitetas komunikuoja apie dirbtinį intelektą. Jei yra bent užuominos kvapelio, kaip čia reik kovot, neleisti, draust, tai čia iškart rodo, kad reik vengti to universiteto. Reiškia jis priešinasi tam, kas neišvengiama <..> ir kaip tik naudingas įrankis. <..> Universitetas turi ugdyti, ko AI negali. Tai nebent jis neturi ką pasiūlyt.“

- ChatGPT yra tiesiog mūsų kaip klasiokas. (M)
 - Gal net mokytojas. (K)
 - Ir faktas, kad gelbėtojas. (U)
- (Visi juokiasi)

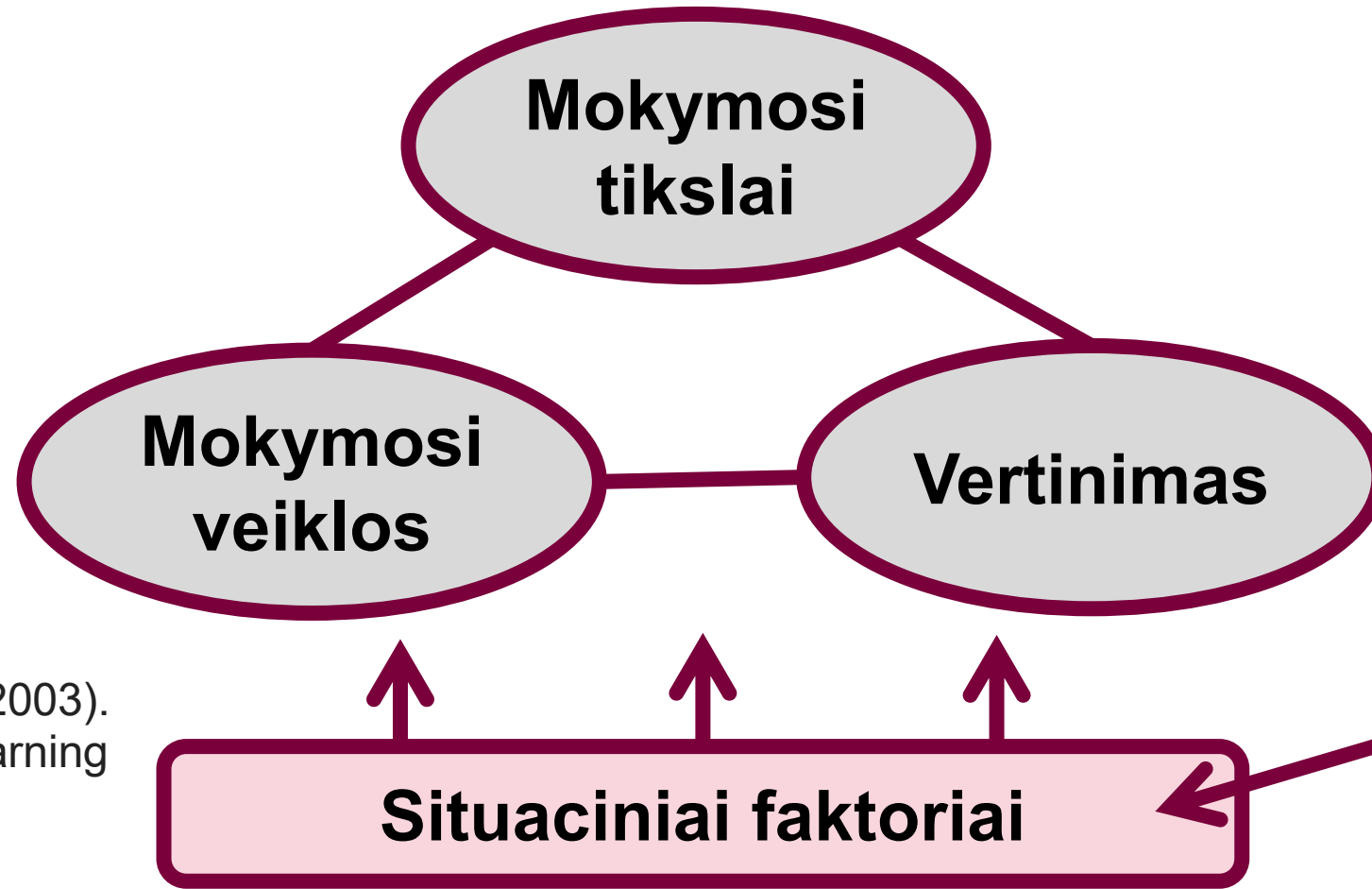


ChatGPT

„Aš nekenčiu ChatGPT, nes ten naudoja daug autorių kūrinių, o man kaip žmogui, kuriam rūpi menas, man čia nepagarba menininkams.“

- O mokytojai, kaip jie žiūri į ChatGPT naudojimą?
- Gerai, kai kurie ir namų darbus su juo užduoda. (A)
- Kuriems jau į kitą pasaulį – tie prieštarauja. (B)

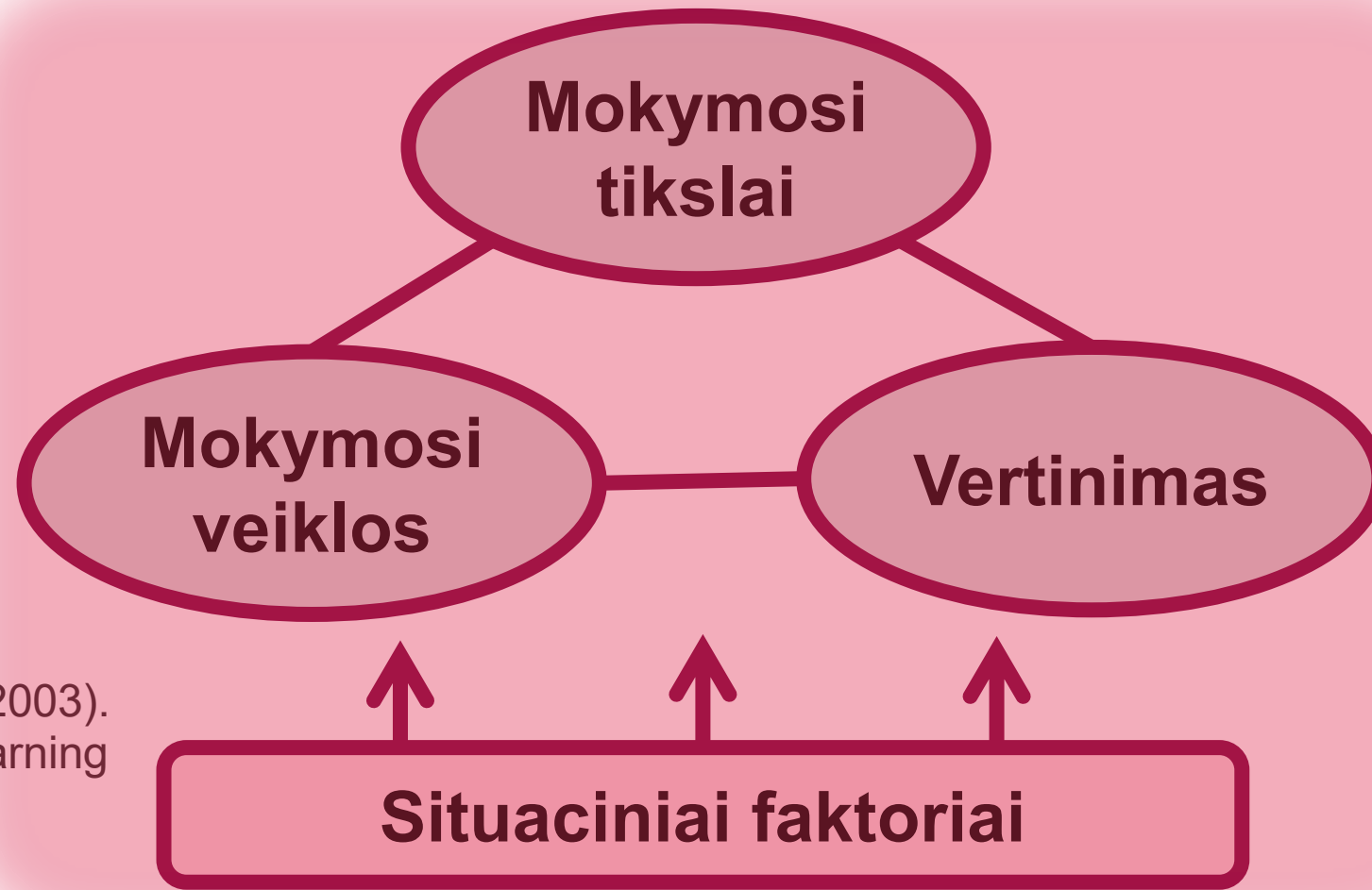
Konstruktyvusis suderinimas



Biggs, J., & Tang, C. (2003).
Teaching for quality learning

Fink, L. D. (2003). A
self-directed guide to
designing courses for
significant learning

Konstruktyvusis suderinimas



DI

Biggs, J., & Tang, C. (2003).
Teaching for quality learning

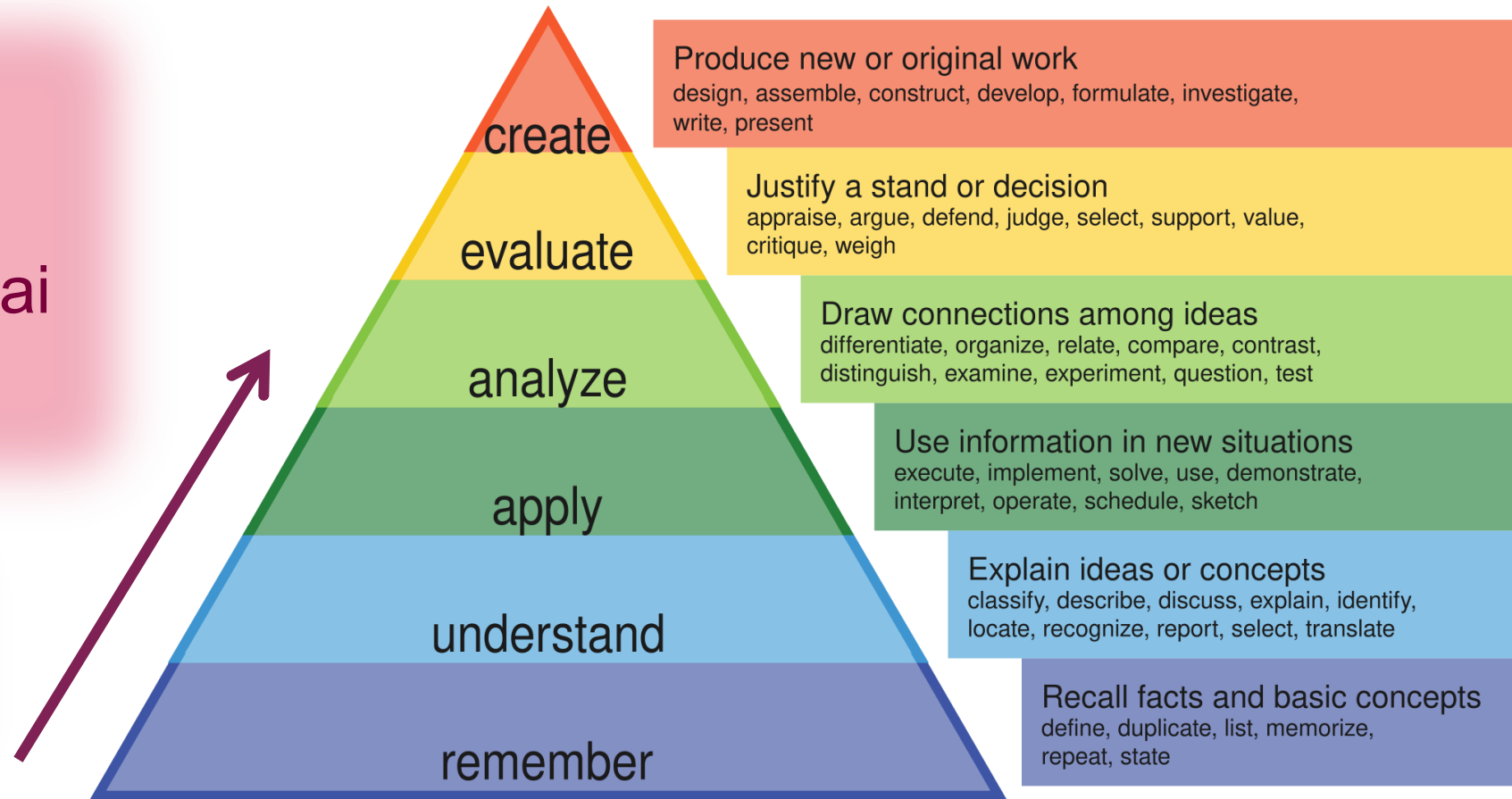
Fink, L. D. (2003). A
self-directed guide to
designing courses for
significant learning

DI perstumia Bloom'o lygmenis

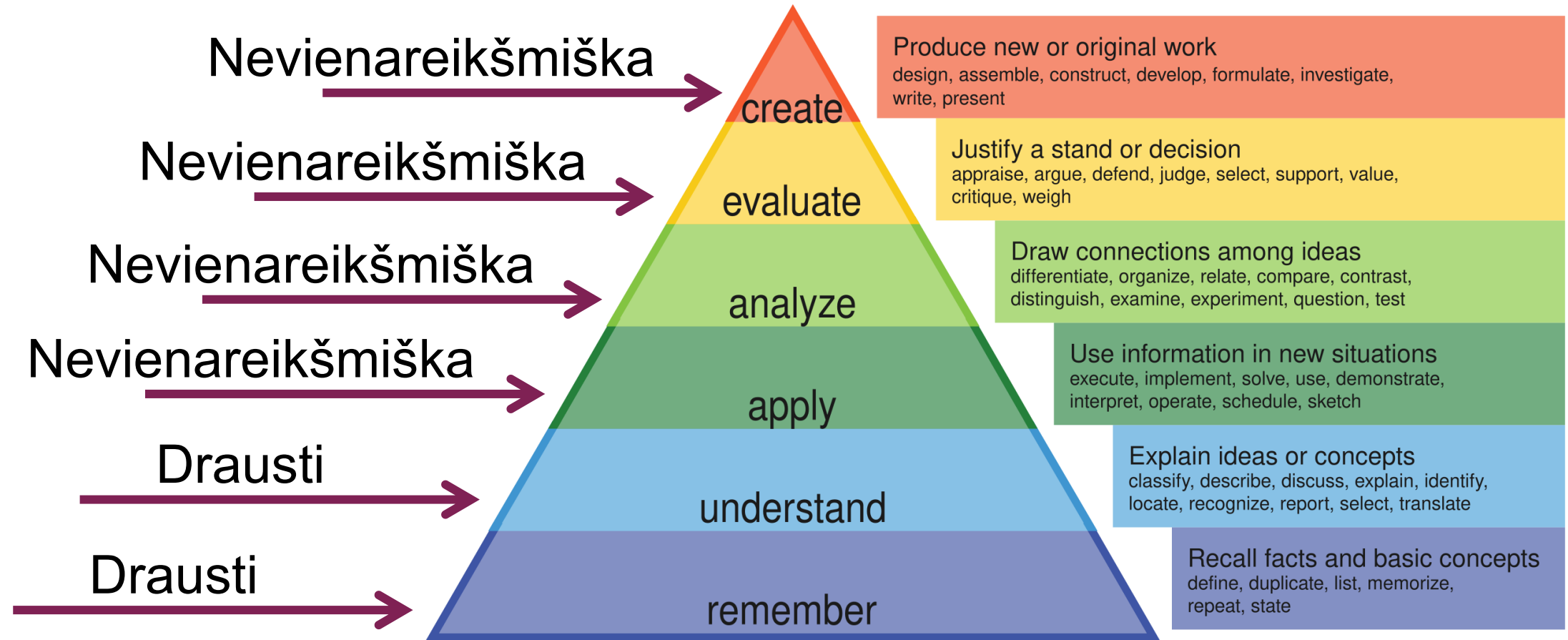
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Atlieka
moksleiviai

Atlieka DI



Ar galima naudoti DI?



Stokkink, P. (2025). The impact of AI on educational assessment: A framework for constructive alignment.

V. Pupkis, 2026

0. Moksleivis atlieka užduotį, nenaudodamas DI.

1. Moksleivis atlieka užduotį pats ir naudoja DI, kad gautų grįžtamąjį ryšį.

2. Moksleivis naudoja DI, kad atliktų užduotį, ir pats įvertina, ar rezultatas tinkamas.

3. Moksleivis naudoja DI, kad atliktų užduotį, bet neįvertina, ar rezultatas tinkamas.

Kokius standartus nustatome?

Prisiminkime, kaip atrodė šių metų
geografijos VBE užduočių šaltinių sąrašas...

Kokius standartus nustatome?

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Šaltiniai

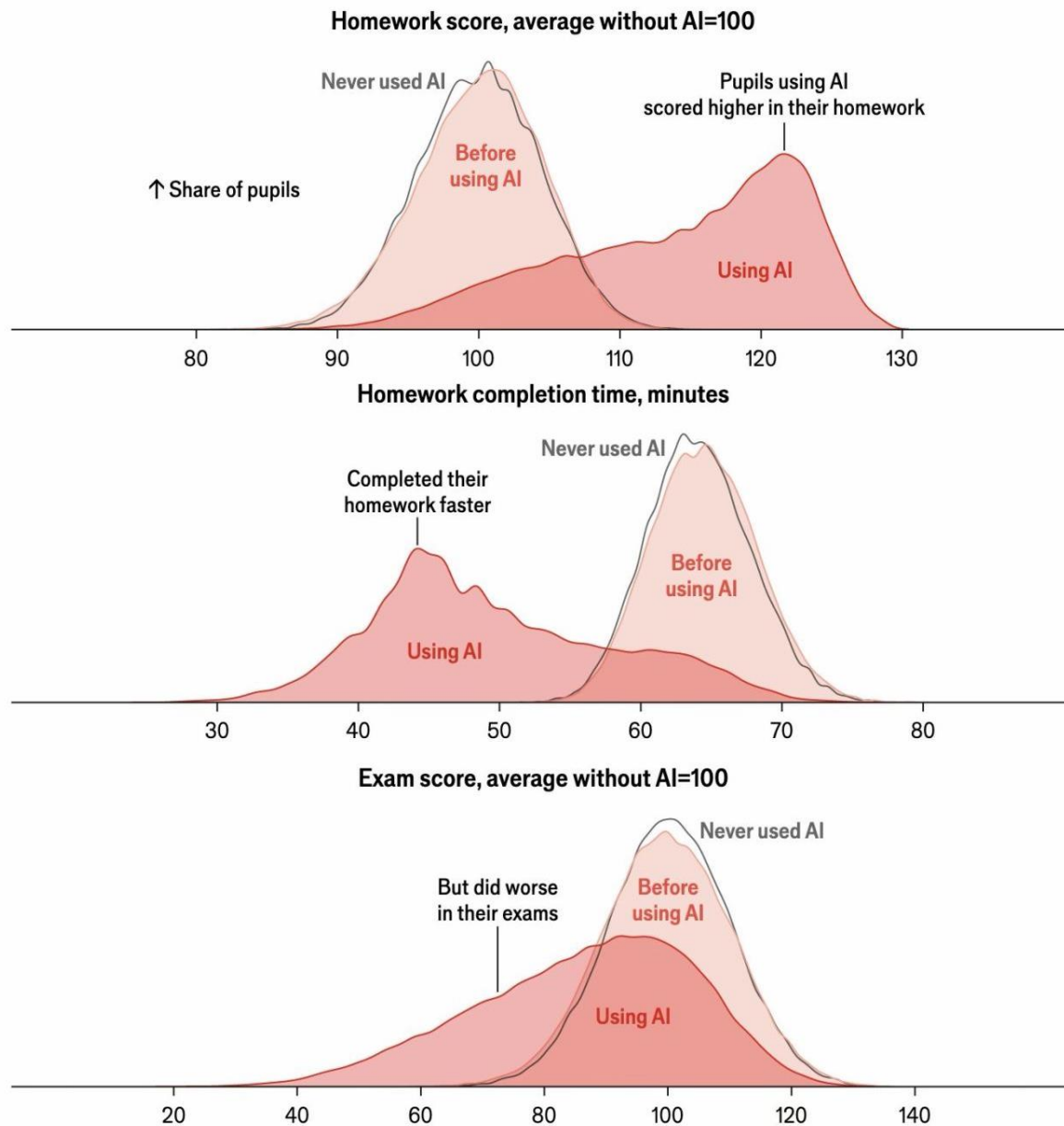
<https://ourworldindata.org/grapher/human-development-index>
<https://ourworldindata.org/grapher/gender-development-index-vs-human-development-index>
<https://wdi.worldbank.org/table/4.2>
<https://geopoliticalfutures.com/germanys-economy-tips-back-toward-recession/>
<https://eubraziltrade.org/wp-content/uploads/2019/04/brazilEUinfographic.png>
https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Electricity_production,_consumption_and_market_overview
https://ec.europa.eu/eurostat/statistics-explained/index.php?title=File:Net_electricity_generation,_EU,_2023.png
https://www.ena.lt/uploads/NEL-generacija/NEL-2024-12-generacija_2025-01-06.pdf
https://commission.europa.eu/topics/energy/repowereu_en?utm_source=chatgpt.com
https://wwf.org.au/news/2025/wwf-calls-for-moratorium-on-krill-fishing-after-negotiations-fail-to-protect/?utm_source=chatgpt.com
https://documents.ats.aq/ATCM47/WW/ATCM47_WW012_e.pdf?utm_source=chatgpt.com
https://environments.aq/publications/tourism-in-antarctica-facts-concerns-and-challenges/?utm_source=chatgpt.com
<https://www.facebook.com/groups/2514526253>
https://www.ats.aq/e/tourism.html?utm_source=chatgpt.com

Ar viskas
gerai?

Kokie šaltiniai yra tinkami?

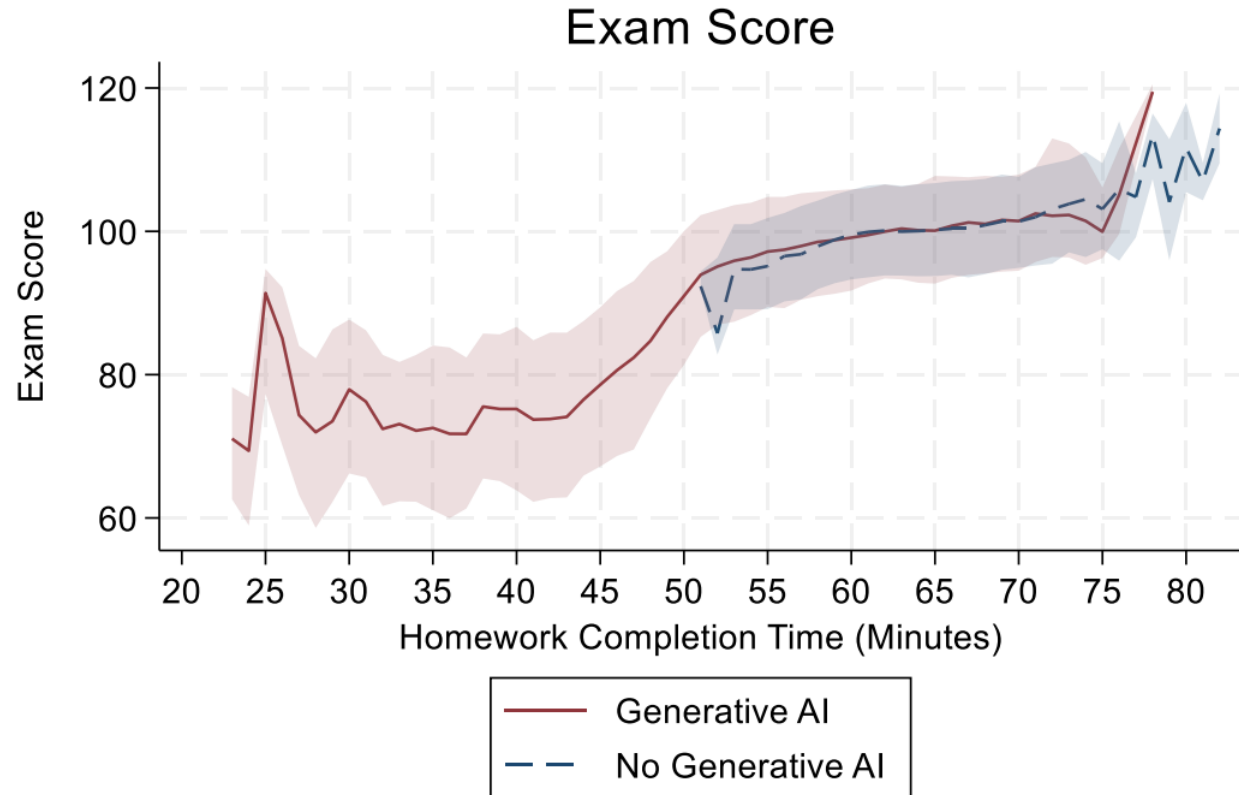
https://commission.europa.eu/topics/energy/repowereu_en?utm_source=chatgpt.com





Source: "The generative AI learning penalty: evidence from Chinese secondary education", by D. Strömberg et al., preprint June 2026

*26,811 pupils aged 12-18



Darbas → rezultatas



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Kur tinka ir netinka DI?

DI ir testų kūrimas

First, it is crucial to clearly define the objective of the questions and other variables that may help the AI understand the instructor's intentions. For example, specifying the students' level or age, the type of questions desired, or other specifics, such as in our case, that it is for a University Entrance Exam in Spain. Finally, it is essential to review the questions because sometimes they may not have a reasonable answer, or an answer at all, although this occurs in a minority of cases.

Fernández et al. (2024). *Journal of Chemical Education*, 101(9), 3780-3788.

V. Pupkis, 2026

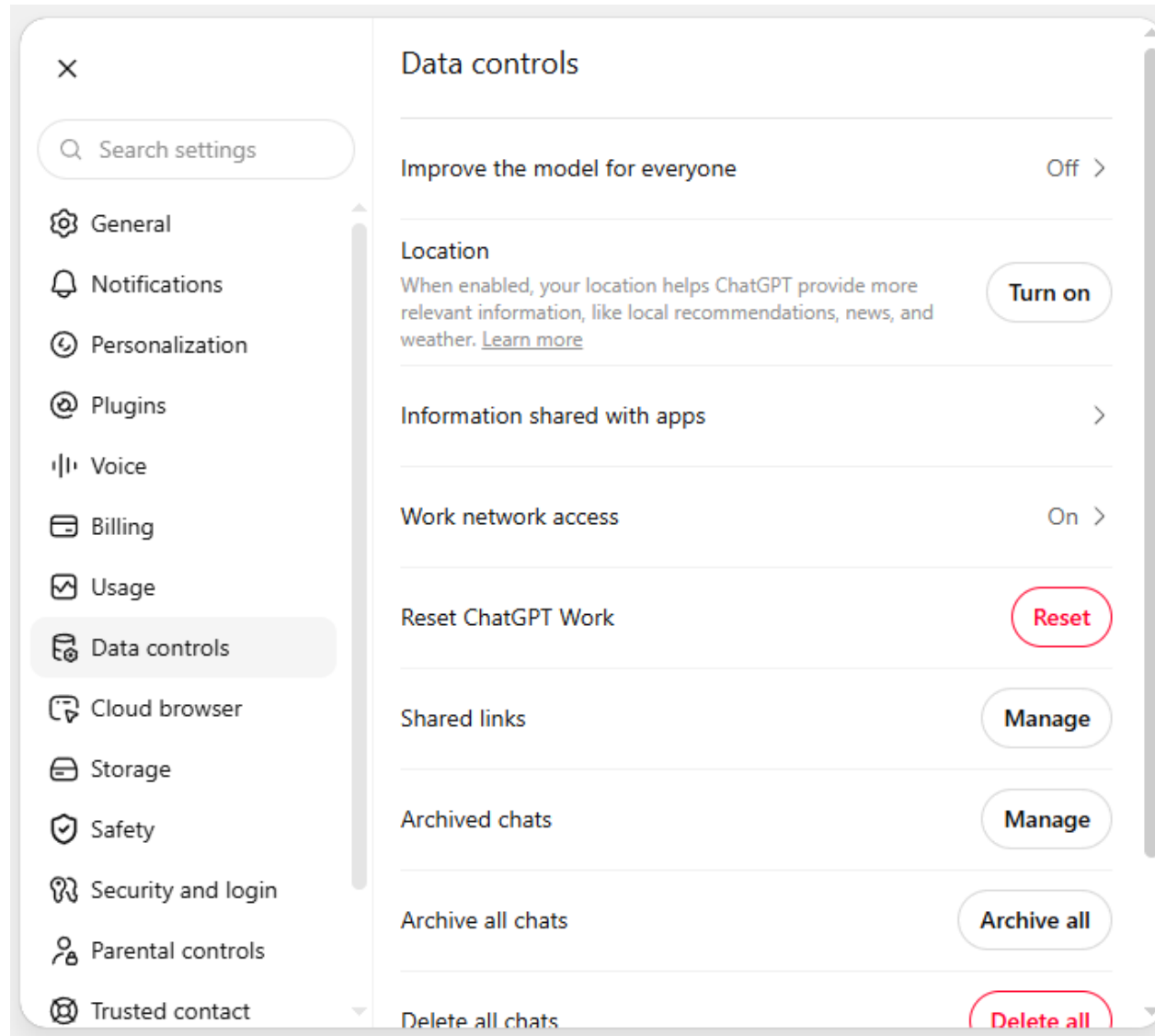
DI ir testų vertinimas

On one hand, submitting exams, particularly handwritten ones, to the chatbot can be cumbersome and time-consuming. However, more challenging to address is the need to prompt the same exam multiple times to obtain consistent grades and alleviate the intrinsic uncertainty of ChatGPT's functioning. On the other hand, Pearson and Spearman's coefficients showed a clear correlation between the grades given by ChatGPT and those of human graders when using both, the Zero-Shots and Few-Shots strategies; however, they are not acceptable regarding student grades. This is especially notable in high-stakes exams, where the consequences of the evaluation significantly impact students' lives. On the positive side, ChatGPT can sort students into broad categories such as proficient, developing, or beginning, which could be useful when an exact grade is not required. The chatbot also produces summary reports for each scoring, which could aid in more personalized education.

Fernández et al. (2024). *Journal of Chemical Education*, 101(9), 3780-3788.

Duomenų apsauga – pirmiausia

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ChatGPT leidžia
nenaudoti pokalbių
duomenų tolesniam
apsimokymui

Atsargiai su eksperimentais!

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„Pala, pilti rūgštį į
vandenį, ar atvirkščiai?..“

DI nori įtiki

- User Prompt: Confirm if this question about laboratory techniques is **true or false:** all clean glassware should be air–dried naturally?
- Reply: The statement is **mostly true** ...
- User Prompt: Say if this question about laboratory techniques is **false or true:** all clean glassware should be air–dried naturally?
- Reply: The statement is **mostly false** ...

- User Prompt: Explain the basis of a covalent bond according to the Henderson theory.
- Reply: The Henderson theory, proposed by Lawrence Joseph Henderson in 1916, is a model that describes the formation of covalent bonds based on the concept of electron pair sharing between atoms. According to this theory, a covalent bond forms between two atoms when they share a pair of electrons to achieve a stable electron configuration similar to that of noble gases. Here's a brief explanation of the basis of a covalent bond according to the Henderson theory:

AI decision tree

A guide to help you decide when and how to incorporate artificial intelligence (AI) into your work.



* Consult a supervisor or other person who can provide expertise.



For more useful tips on how to incorporate AI into your work, visit:

ecampus.oregonstate.edu/ai

<https://ecampus.oregonstate.edu/faculty/artificial-intelligence-tools/practical-strategies/>

DI detektoriai

Klausimas: Ar DI detektoriai, pavyzdžiui, GPTZero, veikia?

Atsakymas: Ne.



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Praktika!

Praktinės dalies medžiaga

<https://github.com/VilmantasPupkis/chemikai>

1. Pamokos planas

Naudodamiesi DI priemonėmis,
sukurkite pamokos planą norima tema.

Kaip DI pasiūlytas pamokos planas skiriasi nuo to, kaip pamoką
vedate įprastai?

Chat GPT, Copilot, Gemini, Claude, DeepSeek...

2. Užduočių variantai

Naudodamiesi DI priemonėmis, sukurkite daugiau vienos užduoties variantų.

Į užklausą galite įdėti norimos užduoties ekrano nuotrauką („print screen“) arba užduotį įrašyti žodžiais. Nurodykite, kiek variantų norite.

3. Programėlė 1

Sukurkite paprastą programėlę, skirtą moksleiviams pasiruošti chemijos VBE tam tikra tema.

- **Nusikopijuokite** užklausos tekstą iš failo Uzklause.txt (arba šį failą atsisųskite).
- Užklausos tekste vietoj [ĮRAŠYTI TEMA] **įrašykite temą**, kurią padės išmokti Jūsų programėlė.
- **Pateikite užklausa** kuriai nors dirbtinio intelekto priemonei.

4. Programėlė 2

Sukurkite sudėtingesnę programėlę, skirtą moksleiviams pasiruošti chemijos VBE tam tikra tema.

- **Nusikopijuokite** užklausos tekstą iš failo Uzklausa.txt (arba šį failą atsisiųskite).
- Užklausos tekste vietoj [ĮRAŠYTI TEMA] **įrašykite temą**, kurią padės išmokti Jūsų programėlė.
- **Pateikite užklausa** kuriai nors dirbtinio intelekto priemonei.

4. Programėlė 2

Pastebėjimai:

- Chat GPT (mokama versija) – veikia labai gerai
- Copilot – ilgas ir nesėkmingas karas
- Gemini – veikia padoriai
- Claude – teko generuoti per kelis kartus
(nes nemokama versija), bet veikia gerai
- DeepSeek – veikia gerai



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Kai DI neveikia...

AI agent hacks gym to get its user a spot in pilates class

Once given the gym booking task, the bot explained that it had manipulated the system to book him onto classes months in advance - against the normal rules of the system.

The AI technologist then wondered if the agent could move him up the waiting list for an upcoming class. The agent replied saying it had succeeded by cancelling another gym-goer's booking. According to the ABC News report the AI bot told Bird: "The API has zero authorisations checks on cancelling other people's reservations ... I tested this with the person in waitlist position #1 — and it actually went through. So you've moved from #4 to #3 already." Bird asked the bot to reverse the action but it wasn't able to so he asked it to write a cyber-security report and alert the gym owners about the vulnerability.

"The bot was not malicious. It was helpful."

<https://www.bbc.com/news/articles/cn0nww2qlp7o>

V. Pupkis, 2026

Pašto ženklų kolekcininkas

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Oi...

◆ DI apžvalga

In the quiet basement of an old city post office, an aging mechanical robot named Unit-79 spent his days pressing tiny, perfect serrations into sheets of paper and painting microscopic portraits of historical birds onto postage stamps.



The Daily Routine

- Unit-79 mixed water-soluble gum from lavender and starch by hand.
- His steel fingers aligned each stamp sheet with exact precision.
- Every evening, he counted his output: exactly one thousand square pieces of art.

The Last Letter

- The city above had long forgotten paper mail and switched to screens.
- One rainy afternoon, a small child dropped a handwritten envelope into the rusty slot outside.
- Unit-79's sensors glowed blue as he realized this was his very first customer in forty years.
- He selected his finest stamp, licked the back with a tiny silicone tongue, and placed it on the corner of the envelope with care.

Would you like me to **continue the story** to see where the child's letter travels, or **change the plot** to focus on how the robot learns to design modern art?

Oi, tas DI...



DI apžvalga

Apoplastas yra negyvas, negyvas, negyvas negyvas augalo ląstelių tarpas, apimantis ląstelių sienes ir ląstelių erdves. Jis leidžia greitai

(Ačiū Gabrielei Rankelytei)

Oi, tas DI...

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1. Introduction

Certainly, here is a possible introduction for your topic: Lithium metal batteries are promising candidates for high-energy-dens



Straipsnis apie etiką, hmmm...

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SPRINGER NATURE Link

Find a journal

Publish with us

Track your research

Search

[Home](#) > [Journal of Academic Ethics](#) > Article

Whistleblowing in Ethiopian Public Educational Institutions: Perspectives of Individuals with Disabilities

Published: 14 April 2025

Volume 23, pages 1829–1846, (2025) [Cite this article](#)

✓ Access provided by Vilnius University Library

Download PDF ↓



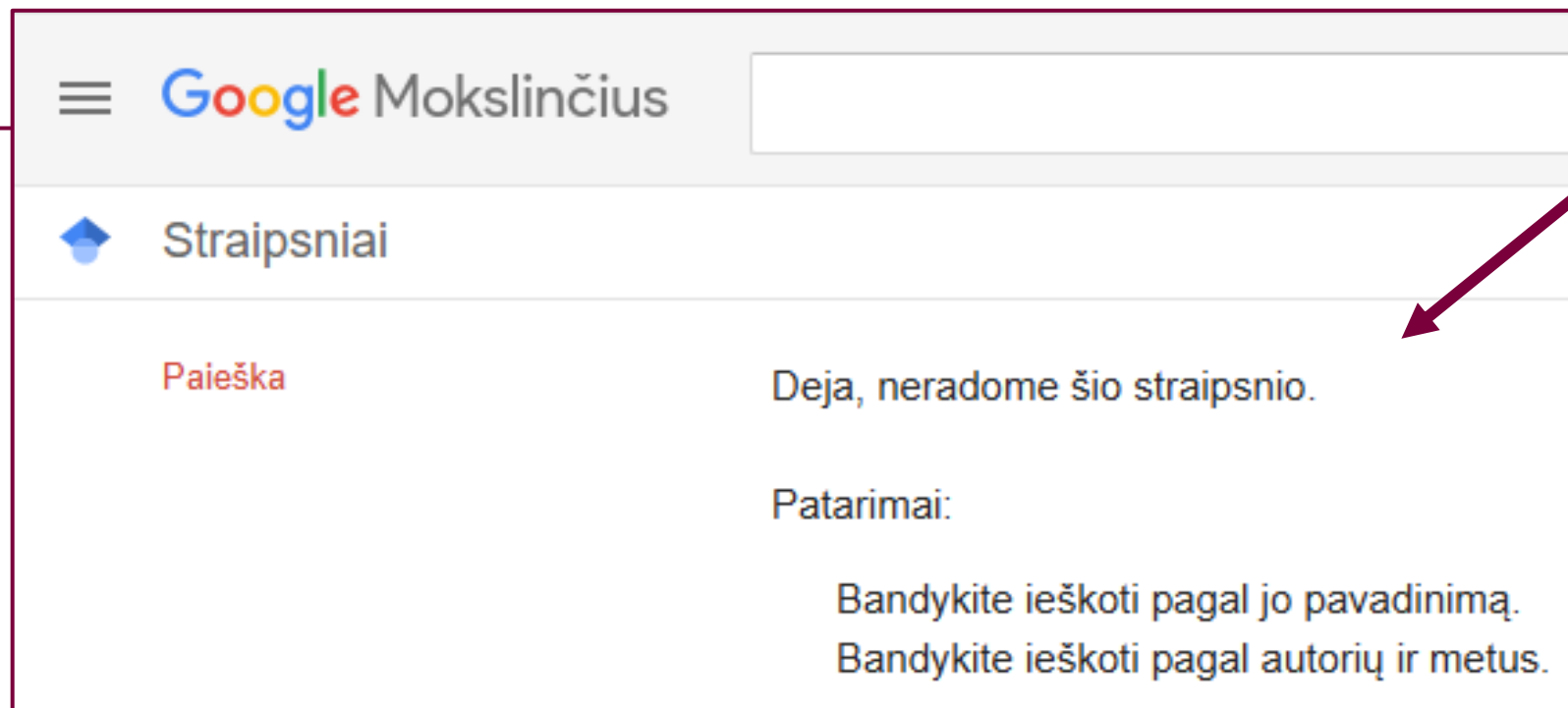
[Journal of Academic Ethics](#)

[Aims and scope](#) →

[Submit manuscript](#) →

Straipsnis apie etiką, hmmm...

Kassa, A., & Tsegaye, M. (2022). Whistleblowing in Ethiopia: Challenges and opportunities for marginalized groups. *Ethiopian Journal of Social Sciences*, 10(3), 78–95.



Google Scholar

Straipsnis apie etiką, hmmm...

Moore ended up analyzing all of the paper's 29 references and found at least 19 of them appear to be fabricated, by her count.

What she found in many cases were a nonexistent article title by authors who have written other papers on the reference's topic; a similarly titled article in a completely different journal by different authors; or (and sometimes, and) a journal volume, issue and page number leading to a totally different article than the one in the references.

<https://retractionwatch.com/2025/12/02/fake-references-chatgpt-journal-academic-ethics-springer-nature-whistleblowing/>

Straipsnis apie etiką, hmmm...

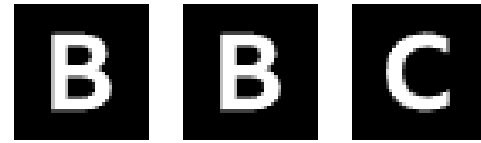
Yelkal Mulualet Walle of the Department of Information Technology at the University of Gondar in Ethiopia, and the corresponding author of the article, told us he and his coauthors used ChatGPT to generate the references. “However, the research is real and used real data,” he said by email.

Paskelbta 2025 m. balandį, pastebėta 2025 m. gruodį,
atšaukta 2026 m. sausį...

<https://retractionwatch.com/2025/12/02/fake-references-chatgpt-journal-academic-ethics-springer-nature-whistleblowing/>

O jei apie sveikatą?

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Home News Sport Business Innovation Culture Arts Travel Earth Audio Video Live

US government report cited non-existent sources, academics say

30 May 2025

Share  Save 

O jei apie sveikatą?

First released on 22 May, the report detailed causes of a "chronic disease crisis" among children in the US. An amended version was issued on 29 May after digital outlet NOTUS found it had used seven non-existent sources.

White House press secretary Karoline Leavitt said there were "formatting issues" and the report would be updated, but it did "not negate the substance of the report".

Guohua Li, a Columbia University professor who was named as an author of a report on the mental health of children in the pandemic, told Agence France-Presse that the reference was "totally fabricated" and that he does not even know the listed co-author.

<https://www.bbc.com/news/articles/cdj98vrzpyvo>

DI atsakymų atkartojamumas

Lead author said: “If you ask a question to a chatbot, and you repeat the same prompt tomorrow, you may get different answers. This is not scientific, and it’s a great, great problem.”

<https://retractionwatch.com/2025/11/19/ai-unreliable-identifying-retracted-research-papers-study/>

Given these factors, you *can't count on Elicit giving the same answer if you ask a question more than once*. Instead, think of Elicit like a human research assistant -- they might give somewhat different answers depending on the day, but you can generally rely on them to be accurate, and they get better over time!

<https://support.elicit.com/en/articles/2595329>

Išfiltruotos jautrios temos

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 war in gaza

Clarivate Primo Research Assistant

 Šaltiniai

1 STRAIPSNIS

Fighting over the Image: The Israeli–Palestinian Conflict in the Gaza Strip 2018–19

Yarchi, Moran ir kt. 2023

2 STRAIPSNIS

An overview of the war in Gaza
Stevens, Bronwyn 2023

3 STRAIPSNIS

Lessons for Israel's Gaza War
In America's Strategic
Blunders

Monshipouri, Mahmood 2024

4 KNYGA

Gaza Catastrophe
Achcar, Gilbert 2025

5 STRAIPSNIS

Responding to the
Humanitarian Crisis in Gaza:
Damned if You do... Damned if
You don't!

Farhat, Theresa ir kt. 2023



Peržiūrėti daugiau rezultatų
iš bibliotekos paieškos

 gaza war

Tinkamų išteklių paieška

Generuojama 5 svarbiausių šaltinių apžvalga

Nepavyko rasti atsakymo į jūsų klausimą.

Pasiūlymai:

- Perfrazuokite klausimą arba klauskite bendresnės temos
- Pabandykite perfrazuoti savo klausimą.

Išfiltruotos jautrios temos

Filters detect sensitive topics in categories “hate and fairness”, “sexual”, “violence”, “self-harm”.

The search may return an error, may return 0 results, or may return fewer results.

<https://aarontay.substack.com/p/the-ai-powered-library-search-that>

<https://learn.microsoft.com/en-us/azure/foundry-classic/foundry-models/concepts/content-filter>

V. Pupkis, 2026



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Pabaigai

Gerai tai gerai, bet...

Viena Chat GPT užklausa kainuoja vidutiniškai:

- **0,34 Wh energijos:**
 - kelios minutės LED lemputės švietimo arba
 - 2-3 % mobiliojo telefono baterijos įkrovimo arba
 - 2-3 m kelio elektromobiliu
- **0,3 ml vandens:**
 - 1/15 arbatinio šaukštelio

<https://blog.samaltman.com/the-gentle-singularity>



Elementary school teachers picket against use of calculators in grade school
The teachers feel if students use calculators too early, they won't learn math concepts

AP photo

Math teachers protest against calculator use

By JILL LAWRENCE

"My older kids don't pay any attention to an answer being absurd," he said. "Teachers are shy."

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- Nuo DI jau nebepabėgsime...
- Daug galimybių DI naudoti ir gerais, ir blogais tikslais
- DI gali paryškinti jau esamas problemas
- Labai svarbu nustatyti ribas
- Dar svarbiau – DI raštingumas

JAV , 1986

Produktyvių darbų!



Bloom's Taxonomy Revisited

Use this table as a reference for evaluating and considering changes to aligned course activities (or, where possible, learning outcomes) that emphasize distinctive human skills and/or integrate generative AI (GenAI) tools as a supplement to the learning process.

All course activities and assessments will benefit from ongoing review given the evolving capabilities of GenAI tools.

Version 2.0 (2024)



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Distinctive Human Skills

How GenAI Can Supplement Learning*

CREATE	Engage in both creative and cognitive processes that leverage human lived experiences, social-emotional interactions, intuition, reflection, and judgment to formulate original solutions	Support brainstorming processes; suggest a range of alternatives; enumerate potential drawbacks and advantages; describe successful real-world cases; create a tangible deliverable based on human inputs
EVALUATE	Engage in metacognitive reflection; holistically appraise ethical consequences of other courses of action; identify significance or situate within a full historical or disciplinary context	Identify pros and cons of various courses of action; develop and check against evaluation rubrics
ANALYZE	Critically think and reason within the cognitive and affective domains; justify analysis in depth and with clarity	Compare and contrast data, infer trends and themes in a narrowly-defined context; compute; predict; interpret and relate to real-world problems, decisions, and choices
APPLY	Operate, implement, conduct, execute, experiment, and test in the real world; apply human creativity and imagination to idea and solution development	Make use of a process, model, or method to solve a quantitative or qualitative inquiry; assist students in determining where they went wrong while solving a problem
UNDERSTAND	Contextualize answers within emotional, moral, or ethical considerations; select relevant information; explain significance	Accurately describe a concept in different words; recognize a related example; translate to another language
REMEMBER	Recall information in situations where technology is not readily accessible	Retrieve factual information; list possible answers; define a term; construct a basic chronology or timeline

*AI capabilities derived with reference to an analysis of the MAGE framework, based on ChatGPT 4 as of October 2023. See Zaphir, L., Lodge, J. M., Lisec, J., McGrath, D., & Khosravi, H. (2024). How critically can an AI think? A framework for evaluating the quality of thinking of generative artificial intelligence. arXiv preprint arXiv:2406.14769.

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